Bangladesh University of Engineering and Technology

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EEE 206

Project Title:

SINGLE PHASE CAPACITOR RUN MOTOR

GROUP: 04

Submitted to

Ramit Dutta

Lecturer, Department of Electrical and Electronics Engineering

Ashikur Rahman Jowel

Lecturer, Department of Electrical and Electronics Engineering

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Date of Submission: 11 September 2023

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EEE 206

Energy Conversion Laboratory

Final Project Report

Section: B2 Group: 04

Capacitor Run Motor

Course Instructors:

Ramit Dutta

Lecturer, Department of Electrical and Electronics Engineering

Ashikur Rahman Jowel

Lecturer, Department of Electrical and Electronics Engineering

Signature of Instructor: ______________________________

Academic Honesty Statement:

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

Signature: Full name-Student Id

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Forwarding letter

Date: 11 September 2023

To

Ramit Dutta (Lecturer)

Ashikur Rahman Jowel (Lecturer)

Subject: Report Submission on the project "Capacitor Run Motor".

Dear Sir,

With sincere respect and profound appreciation, we are pleased to submit our report on the project titled "Capacitor Run Motor" for your evaluation and consideration. We extend our heartfelt thanks for entrusting us with this opportunity and for your invaluable guidance throughout this undertaking.

Our primary objective in this project was to design and implement an efficient capacitor-run motor, carefully considering factors such as cost-effectiveness and the availability of materials in the market. We were equally committed to the ethical principles of our field, ensuring that our work adhered to high standards of integrity and authenticity.

In presenting this report, we acknowledge the possibility of inadvertent errors, typographical mistakes, or editing oversights, and we take full responsibility for addressing any such issues.

We are profoundly grateful for the chance to translate theoretical knowledge into practical application. This project has provided us with a unique platform to bridge the gap between classroom learning and real-world engineering.

We earnestly hope that our endeavors align with the core objectives of this project and meet the standards set by our academic institution.

Once again, we express our sincere thanks for your mentorship and guidance, and we eagerly await your feedback and assessment of our work.

Sincerely,

Team members of group 4

Tanvir Arafat Fahim	2006118
Fabliha Labiba	2006119
Md. Raihan Siddik	2006120
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Contents:

Project Objectives:

Conversion of Capacitor Start Motor: The primary objective of this project is to convert a standard capacitor start motor into a capacitor-run motor. This conversion process will involve selecting the appropriate capacitor, designing the necessary circuit modifications, and implementing them effectively to enable the motor to run with improved efficiency.

Demonstrate Practical Understanding: To demonstrate a practical understanding of the principles and theories related to capacitor-run motors, emphasizing the differences in operation, construction, and applications.

Efficiency Enhancement: To assess and improve the motor's efficiency and power factor through the addition of the capacitor, ensuring that the motor operates optimally for specific applications, such as fans and pumps.

Troubleshooting and Optimization: To develop troubleshooting skills by identifying and addressing any issues or challenges that may arise during the conversion process.

Safety Protocol Adherence: To prioritize safety throughout the project by implementing appropriate safety measures and precautions during the conversion, testing, and operation of the motor.

Introduction:

Our project embarks on a journey to delve into the fundamental principles behind the design and construction of a capacitor-run motor. The capacitor-run motor, a member of the alternating current machine family, operates by harnessing the power of electromagnetic induction, generating the necessary current and flux within its rotor to generate torque.

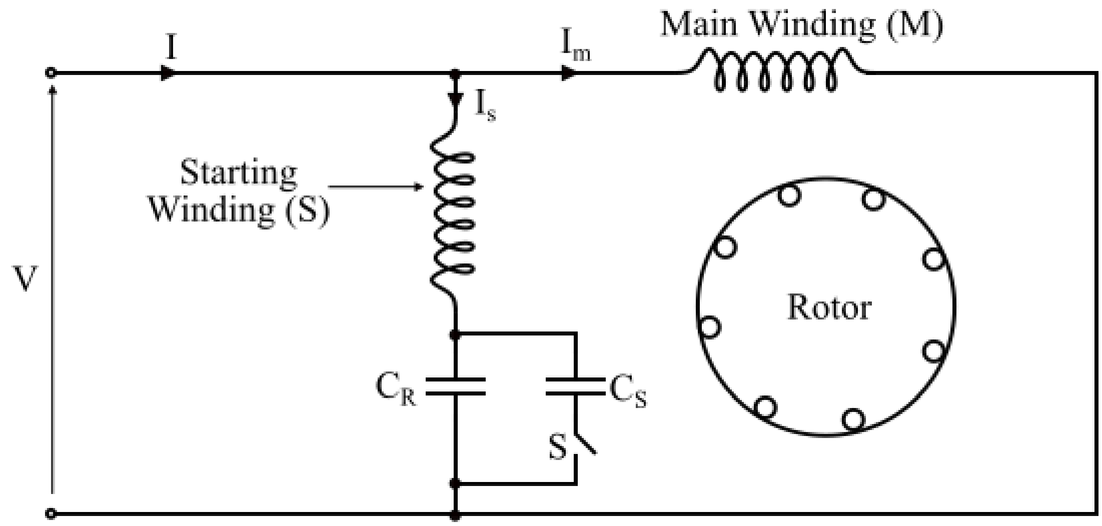
Throughout the course of this project, we will embark on an exploratory mission, dissecting the intricate components that constitute the anatomy of these motors. Our aim is to shed light on the essential considerations and methodologies intrinsic to motor design. This journey will involve gaining a profound understanding of the governing principles dictating its operation, the pivotal role played by individual components and parameters in determining its performance characteristics, the judicious selection of materials, and the optimization of design choices from a cost perspective.

As we navigate the intricacies of the design process, we will not only uncover the core concepts underpinning the operation of capacitor run motors but also engage in a thorough discussion of critical design considerations. Moreover, we will illustrate how to bridge the chasm between theoretical knowledge and practical engineering solutions, emphasizing the significance of this translation in the realm of electrical and electronic engineering.

Design Analysis:

Theory:

The capacitor-start capacitor-run motor is a type of single-phase induction motor. The capacitor-start capacitor-run induction motor is also known as ***two value capacitor motor***. The schematic diagram of a capacitor-start capacitor run induction motor is shown below.



The capacitor-start capacitor-run induction motor consists of a squirrel cage rotor and its stator has two windings, viz. the **starting** or **auxiliary winding** and the **main** or **running winding**. The two windings are displaced by an angle of 90° in the space.

This motor uses two capacitors − the **starting capacitor** *(C_S)* and the **running capacitor** *(C_R)*. The two capacitors are connected in parallel at the instant of starting.

In order to obtain a high starting torque, a large starting current is required. For this, the capacitive reactance in the starting winding should be low.

Since the reactance of the starting capacitor is given by,

$$X_s = \frac{1}{\omega C_s}$$

Hence, for X_S to be small, the value of starting capacitor *(C_S)* should be large. The starting capacitor C_S is a short-time rated electrolytic capacitor.

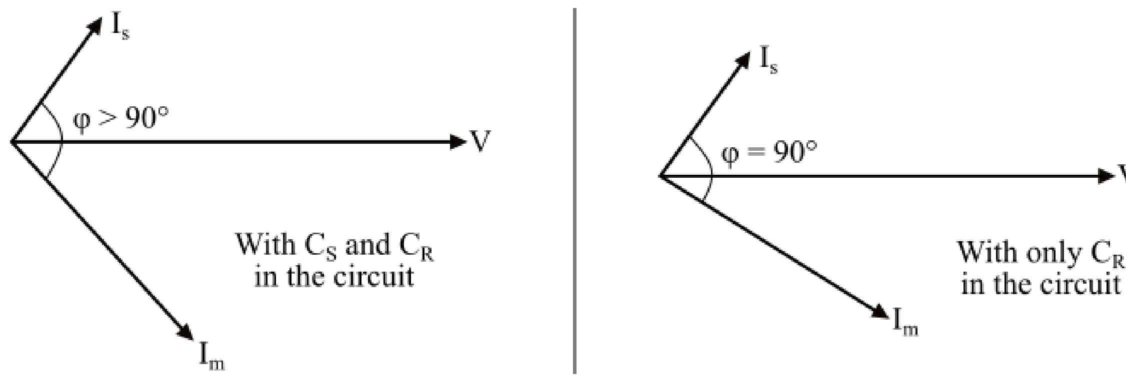
During the normal operation of the motor, the rated line current should be smaller than the starting current. Therefore, the capacitive reactance of the running capacitor should be high and is given by,

$$X_s = \frac{1}{\omega C_R}$$

Hence, for X_R to be high, the value of the running capacitor *(C_R)* should be small. The running capacitor is a long-time rated capacitor and is usually of oil-filled paper construction.

As the motor attains the normal speed, the starting capacitor *(C_S)* is disconnected from the motor circuit by a centrifugal switch (S) and the running capacitor *(C_R)* remains permanently connected in the circuit. Since one capacitor *(C_S)* is used only at starting and the other capacitor *(C_R)* for continuous running, the motor is known as *capacitor-start capacitor-run motor*.

The phasor diagram of the capacitor-start capacitor-run motor is shown below. At starting both the capacitors are in the circuit, therefore, the phase angle ϕ is greater than 90°. When the starting capacitor *(C_S)* is disconnected from the circuit, then the phase angle becomes 90° electrical.



Construction:

The construction of a capacitor-run motor, also known as a permanent-split capacitor (PSC) motor, involves several key components and features. Here's a breakdown of the construction elements of a typical capacitor-run motor:

<u>1. Stator:</u> The stator is the stationary part of the motor and consists of laminated steel cores with evenly spaced slots. These slots accommodate the motor windings, which are usually made of copper wire.

<u>2. Main Winding:</u> The main winding is one of the two windings in the stator. It is connected directly to the AC power supply and generates the primary magnetic field in the motor.

<u>3. Auxiliary Winding (Start Winding):</u> The auxiliary winding, also known as the start winding, is the second winding in the stator. It is positioned at a specific angle to the main winding to create a phase difference. The start winding is connected in series with the start capacitor.

<u>4. Run Capacitor:</u> The run capacitor is another electrolytic capacitor, but it is connected in parallel with the auxiliary winding. The run capacitor provides a continuous phase shift while the motor is running, improving efficiency and torque.

<u>5. Rotor:</u> The rotor is the rotating part of the motor and is usually made of iron laminations or a squirrel-cage design, depending on the type of motor. The rotor's movement is induced by the rotating magnetic field generated by the stator windings.

6. Bearings: Bearings are used to support the rotor shaft and allow it to rotate smoothly within the motor housing. High-quality bearings are crucial for motor longevity and performance.

7. Motor Housing: The motor housing, often made of cast iron or aluminum, encases the stator and rotor, providing protection and structural support. It also contains ventilation holes for cooling purposes.

8. End Bells: These are covers on either end of the motor housing, and they house the bearings and support the rotor shaft. They also provide additional protection and mounting points for the motor.

The construction of a capacitor-run motor is designed to provide the necessary phase shift and starting torque, making it suitable for various applications that require single-phase AC motors. The careful arrangement of windings, capacitors, and other components ensures efficient and reliable motor operation.

Diagrams:

Casing with stator slot

Capacitor

Stator with main winding and auxiliary winding.

Cooling Fan

End Bells, Terminal box, Rotor shaft

Rotor

Circuit Diagram:

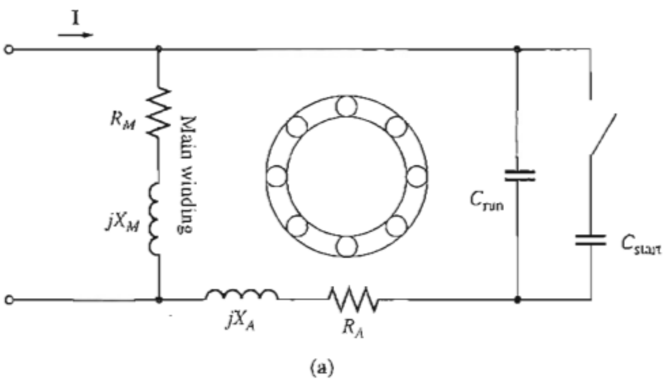


Fig: A capacitor Start Capacitor Run Induction Meter

Working Principle:
A capacitor-run motor operates on the principle of creating a rotating magnetic field. Here's how it works from a magnetic point of view:

1 Start Winding:

The motor has *two windings - the main or run winding and the auxiliary or start winding.* The main winding is responsible for generating the motor's primary magnetic field.

2. **Phase Difference**:

When AC voltage is applied to the motor, it creates an alternating current in both windings. However, there is **a phase difference** between the currents in the main and auxiliary windings due to the capacitor.

3. **Rotating Magnetic Field:**

The phase difference in the currents causes a time-delayed magnetic field in the auxiliary winding. This time delay creates a rotating magnetic field within the motor.

4.Rotor Interaction:

The rotor, which is a core made of iron, is placed within this rotating magnetic field. As the magnetic field rotates, it induces currents in the rotor due to electromagnetic induction. These currents create their own magnetic field.

5 Torque Generation:

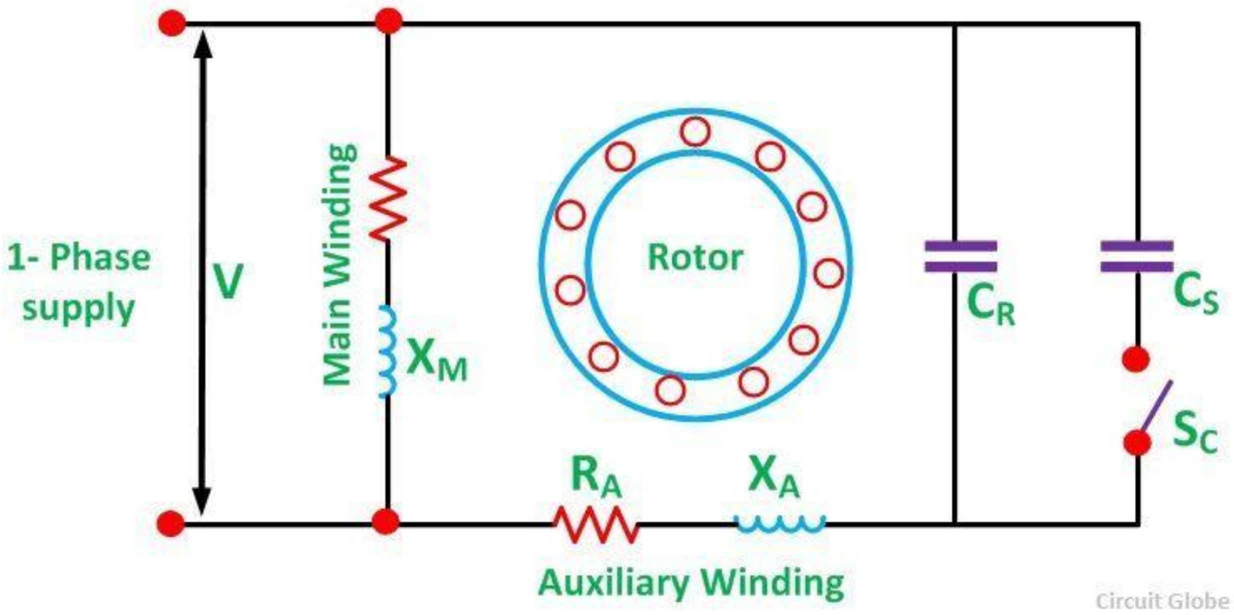
The interaction between the rotating magnetic field produced by the windings and the induced magnetic field in the rotor generates torque. This torque causes the rotor to start rotating.

6. Capacitor's Role:

The capacitor is crucial in creating the phase difference between the currents in the windings. It helps in starting the motor by providing the necessary time delay for the auxiliary winding to create the rotating magnetic field. In summary, a capacitor-run motor utilizes the phase difference created by the capacitor to generate a rotating magnetic field, which in turn induces currents and generates torque in the rotor, causing the motor to start and run efficiently.

FORMULAS

Here's a basic overview with relevant formulas:



1 **Main Winding Current (Im):**

The main winding current is determined by the applied voltage (V) and the main winding impedance (Z_m). It lags the voltage by the angle φ_m due to the motor's inductive nature. Formula: Im = V / Zm, where $Z_m = R_m + jX_m$ (complex impedance of the main winding)

2 Start Winding Current (Is)/Auxilary Winding(Ia) :

The start winding current is determined by the applied voltage (V) and the start winding impedance (Z_s). It also lags the voltage but by a different angle φ_s due to the motor's capacitive nature. Formula: Is = V / Zs, where Z_s = R_s - jX_s (complex impedance of the start winding)

3 Total Current (I_total):

The total current in the motor is the vector sum of the main and start winding currents. *Formula: I_total = √((I_m * cos(φ_m))^2 + (I_m * sin(φ_m) + I_s * cos(φ_s))^2)*

4 **Rotating Magnetic Field Frequency (f):**

The frequency of the rotating magnetic field is determined by the AC power supply frequency (f_supply).

Formula: f = f_supply

5 Phase Difference Angle (φ):

The phase difference between the main and start winding currents is crucial for creating the rotating magnetic field.

Formula: φ = φ_m - φ_s

6 Capacitor Value (C):

The value of the capacitor is essential in creating the phase difference. It can be calculated using the *desired phase shift angle and the motor's characteristics.*

Formula: $C = 1 / (2\pi f_{supply} * R_s * \tan(\varphi))$

7 Torque (T):

The torque produced by the motor *is proportional to the product of the main winding current, start winding current, and the sine of the phase difference angle.*

Formula: $T \propto I_m * I_s * \sin(\varphi)$

These formulas provide a simplified understanding of how a capacitor-run motor works. In practice, motor design involves more complex considerations, such as motor geometry, core materials, and efficiency, but these equations capture the fundamental principles of generating a rotating magnetic field and starting the motor using a capacitor.

Nameplate/ Rated Values:

Type	JY7134
Power	370 W
Voltage	220 V
Current	4.2 A
Frequency	50 Hz
Insulation	Positive
Speed	1400 rpm


单相异步电动机
型号 JY7134 编号
功率 370W 电压 220V 电流 4.2A
频率 50Hz 绝缘 E 转速 1400r/min
上海微型电机厂


single phase asynchronous motor
type JY7134 编号
power 370W Voltage 220V Current 4.2A
frequency 50Hz insulation Positive speed 1400r/ming
Shanghai Micro Motor Factory

Execution and Implementation:

Description

The implementation of a Single-Phase Capacitor-Run Motor project is a comprehensive endeavor in electrical engineering, focusing on the creation of an efficient and functional motor system. This project encompasses the design and assembly of electrical circuitry, careful motor construction, capacitor selection for optimized performance, rigorous testing, and safety measures to ensure safe operation. Through meticulous documentation and reporting, it aims to learn and enlighten us, highlighting the motor's wide range of applications, energy efficiency benefits, and its potential for future advancements in electrical engineering and motor technology.

Apparatus and components

A single-phase capacitor-run motor is a type of electric motor commonly used in various applications. It includes several key components that work together to enable its operation. The main components of a single-phase capacitor-run motor are as follows:

Stator: The stator is the stationary part of the motor and consists of a laminated iron core with wire windings. In a single-phase motor, there are typically two windings: the main winding (start winding) and the auxiliary winding (run winding).

Rotor: The rotor is the rotating part of the motor and is located inside the stator. It is responsible for generating mechanical motion when subjected to a magnetic field created by the stator windings.

Capacitor: A capacitor is a key component in a single-phase capacitor-run motor. It is connected in series with the auxiliary winding and is responsible for creating a phase shift in the current between the main winding and auxiliary winding. This phase shift is essential for the motor to start and run efficiently. Here, we used 20µF capacitor.

Insulators: Insulators are used to separate and insulate electrical components from each other to prevent electrical short circuits. They are typically made of materials such as plastic or ceramic and are strategically placed within the motor to maintain electrical isolation.

Wire: Copper wire is used for winding the stator and auxiliary windings. These windings are essential for creating the magnetic fields and conducting electrical current within the motor.

Bearings: Bearings are used to support and allow the rotation of the motor's shaft. Properly lubricated bearings ensure smooth and reliable operation.

Housing: The motor's housing or casing encloses all the internal components and provides protection against dust, moisture, and mechanical damage. It also serves to dissipate heat generated during operation.

These components work together to create the necessary electromagnetic fields, phase shifts, and mechanical motion required for the motor to start and operate efficiently in single-phase capacitor-run motor applications. The specific design and arrangement of these components may vary depending on the motor's size, intended use, and manufacturer.

Work flow

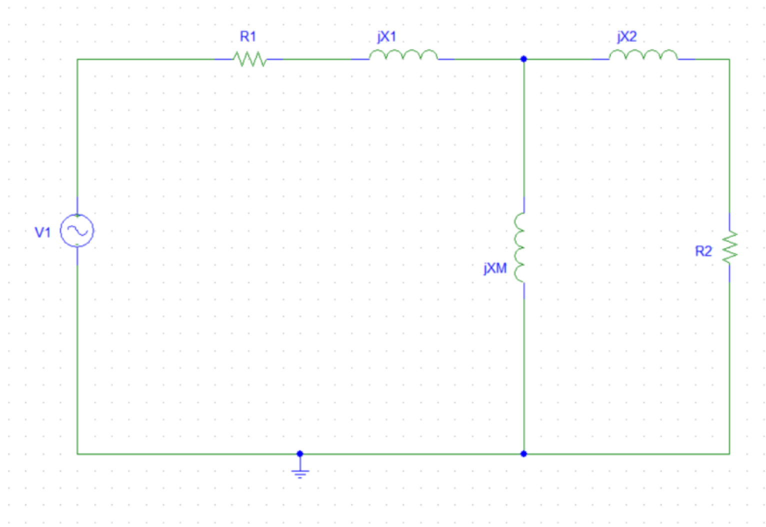
- **Disassembling**
  - **The Capacitor Removal:**
    I. Commenced by ensuring the motor's complete disconnection from the power source to ensure safety.
    II. Locating the capacitor within the motor assembly; typically, it is connected to the electrical circuit.
    III. The wires leading to the capacitor was disconnected, and their positions should be noted for reassembly.
    IV. Capacitor was secured with mounting brackets or clamps and they were removed to release the capacitor from its position.
  - **Housing Detachment:**
    I. The screws, bolts, or fasteners securing the motor housing are typically found on the outer casing.
    II. Once the fasteners were removed, the motor housing can be carefully lifted or slid off to reveal the internal components.
  - **Stator and Rotor Separation:**
    I. With the housing removed, access to the stator and rotor was granted.
    II. The rotor was carefully separated from the stator, care was taken not to damage any components during this process.
    III. The orientation of the rotor was noted, as it will be crucial during reassembly.
  - **Winding Untying (Main and Auxiliary):**
    I. Ties or bindings used to secure the winding coils together were sought on the stator windings.
    II. The ties were carefully untied or removed, with care taken not to damage the winding or its insulation.
    III. The main and auxiliary windings was separated as needed.
  - **Wire Measurement and Turn Counting:**
    I. With the winding untied, the wire diameter was measured using calipers, and the number of turns in the winding coils was counted.
    II. This information is considered valuable for maintenance, repairs, or rewinding.
  - **Insulator Removal:**
    I. Identification of insulators or separators located between the winding coils was done.
    II. Insulators were carefully removed, ensuring that the wire or insulation was not damaged in the process.

- **Assembling**
  - **Safety Precautions:**
    I. Before the assembly process is started, it was ensured that the motor had been completely disconnected from the power source to prevent electrical accidents.
    II. Appropriate safety gear and measures were taken.
  - **Components Preparation:**
    I. It was ensured that all the components, including the stator, rotor, capacitor, insulators, and housing, had been cleaned and they were in good condition.
  - **Insulator Placement:**
    I. The placement of any insulators or separators onto the stator core was initiated. These insulators were used to prevent electrical contact between winding coils.
  - **Windings Preparation:**
    I. Winding coils were untied during disassembly, so it was ensured that they had been properly oriented and organized.
    II. The main and auxiliary windings were aligned in their respective positions on the stator core.
  - **Stator and Rotor Assembly:**
    I. The rotor was carefully positioned within the stator ensuring proper alignment and engagement.
    II. Care was taken not to damage the winding coils or the insulation during this process.
    III. It was ensured that the orientation of the rotor matched the notes or markings made during disassembly.
  - **Windings Re-tying:**
    I. The winding coils were secured with ties or bindings. It was ensured that the ties were snug but not excessively tight to avoid damaging the windings.
    II. It was verified that the winding coils were properly secured to maintain their shape and position.
  - **Capacitor Installation:**
    I. The capacitor was reattached to the electrical circuit, and the wires was connected according to the earlier notes.
    II. The capacitor was secured in its designated location by using any mounting brackets or clamps.

- **Housing Attachment:**
    I. The motor housing was carefully placed over the assembled components, ensuring that it aligned properly with any mounting holes or features.
    II. The housing was secured by tightening the screws, bolts, or fasteners.
- **Electrical Connections:**
    I. The electrical wires were reconnected to their respective terminals.
    II. It was verified that the connections match the earlier notes.
- **Final Inspection:**
    I. A visual inspection of the assembled motor was performed to ensure that all components were placed in their correct positions and that there were no loose parts or wires.
- **Testing:**
    I. Before the motor was returned to service, a test run was conducted to ensure that it operates smoothly and efficiently.
    II. Any unusual noises, vibrations, or overheating was monitored.

Testing and Calculation

When analyzing a single-phase induction motor based on the double-revolving-field theory, we can derive an equivalent circuit for the main winding of the motor when it's operating independently. At a standstill condition, the motor essentially resembles a single-phase transformer with its secondary circuit shorted out. Therefore, the equivalent circuit can be summarized as follows:



Here R_1 and X_1 are the resistance and reactance of the stator winding, X_M is the magnetizing reactance.

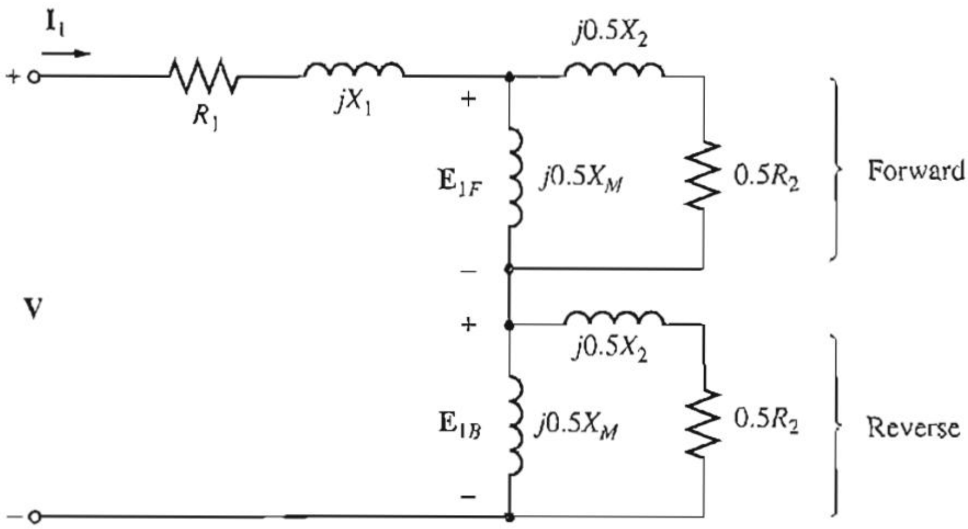
R_2 and X_2 are the referred values of the rotor resistance and reactance.

This motor generates two magnetic fields: a forward magnetic field and a backward magnetic field. Because these fields have equal magnitudes, they both contribute equally to the resistive and reactive voltage drops in the rotor circuit. To simplify our analysis, we can divide the rotor's equivalent circuit into two sections, each representing the effects of one of these magnetic fields.

For the forward magnetic field, the per-unit speed difference between the rotor's speed and the magnetic field's speed is referred to as the slip 's'. Consequently, the rotor resistance in the part of the circuit associated with the forward magnetic field is $0.5R_2/s$.

The forward magnetic field rotates at a speed of n_{sync}, while the reverse magnetic field rotates at a speed of $-n_{sync}$. Therefore, the total per-unit speed difference between the forward and reverse magnetic fields is 2. Since the rotor turns at a speed 's' slower than the forward magnetic field, the total per-unit speed difference between the rotor and the reverse magnetic field is 2 - s. This results in the effective rotor resistance in the part of the circuit associated with the reverse magnetic field being $0.5R_2/(2 - s)$.

So, the final equivalent circuit can be described as follows:



To simplify the calculation of the input current flowing into the motor, it is a common practice to introduce two defined impedances: Z_F, which represents a single impedance that encompasses all the elements associated with the forward magnetic field, and Z_B, which represents a single impedance that encompasses all the elements associated with the backward magnetic field. These impedance values can be expressed as follows:

$$Z_F = R_F + jX_F = \frac{(R_2/s + jX_2)jX_M}{(R_2/s + j\mathrm{X}) + jX_M}$$

$$Z_B = R_B + jX_B = \frac{\left[\frac{R_2}{2-s} + jX_2\right](jX_M)}{\left[\frac{R_2}{2-s} + jX_2\right] + jX_M}$$

In terms of Z_F and Z_B, the current flowing in the induction motor's stator winding is

$$I_1 = \frac{V}{R_2 + jX_1 + 0.5Z_F + 0.5Z_B}$$

Testing data:

DC Test	No Load Test	Blocked Rotor Test
R_1 = 5.8Ω	P = 220W	P = 220W
	I = 1.5A	I = 4.2A
	n_m = 1502 rpm	V= 88V
	V = 220V	

Blocked Rotor Test

P_{SC} = 220W
I_{SC} = 4.2A
V_{SC}= 88V

$R_{SC} = \frac{P_{SC}}{I_{SC}^2} = 12.47\Omega$

$R_1 + R_2 = 12.47\ \Omega$

$Z_{SC} = \frac{V_{SC}}{I_{SC}} = 52.38\Omega$

$X_{SC} = \sqrt{Z_{SC}^2 - I_{SC}^2} = 50.87\Omega$ or, $X_1+X_2 = 50.87\Omega$

For design class D, $X_1 = X_2 = 25.435\Omega$

Slip of the motor is: s = 5.33%

$\frac{n_{sync-n_{nl}}}{n_{sync}} = 0.0533$

$\therefore\ n_{sync} = 1586.56\, rpm \approx 1587\ rpm\quad \frac{120 f_{SE}}{P} = 1587\ rpm$ or, $f_{SE} = 52.9Hz$

$X_{rated} = \frac{f_{rated}}{f_{tested}} \times X_1 = 24.04\Omega$

$X_{1rated} = 24.04\Omega$ and $X_{2rated} = 24.04\Omega$

DC Test

$R_1 = 5.8\Omega$

No Load Test

P_{nl} = 220W
I_{nl} = 1.5A
n_m = 1502 rpm
V_{nl} = 220V

$\cos\theta_{nl} = \frac{P_{nl}}{V_{nl} I_{nl}} = \frac{220}{220\times1.5} = 0.667$

$\therefore\ \theta_{nl} = 48.164°$

$Z_{nl} = \frac{V_{nl}}{I_{nl}} = \frac{220}{1.5} = 146.67\Omega$

$X_{nl} = Z_{nl}\sin\theta_{nl}$ or, $X_{1rated}+\frac{1}{2}(X_{2rated}+X_M) = 109.28\Omega$

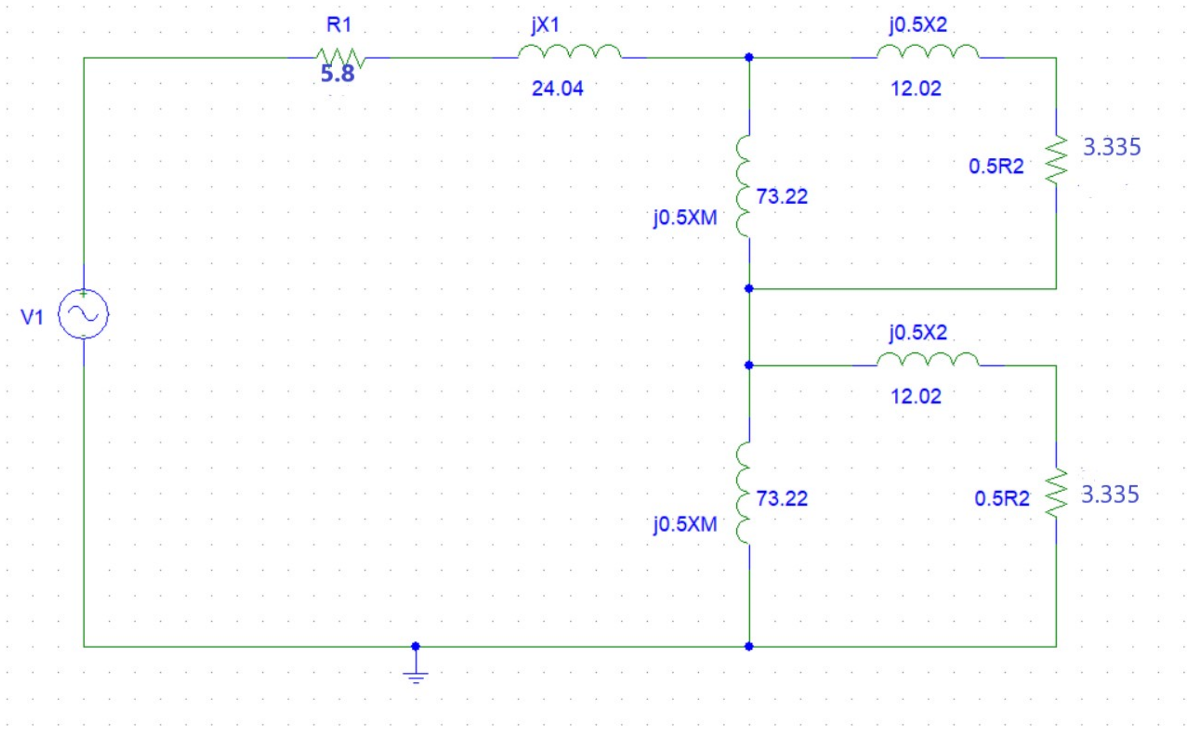
$X_M = 146.44\Omega$ [$X_{1rated} = 24.04\Omega$ and $X_{2rated} = 24.04\Omega$]

$R_{nl} = Z_{nl}\cos\theta_{nl} = 97.83\ \Omega$

$R_1 = 5.8\Omega$

$R_2 = 6.67\Omega$

The equivalent circuit of the motor is:



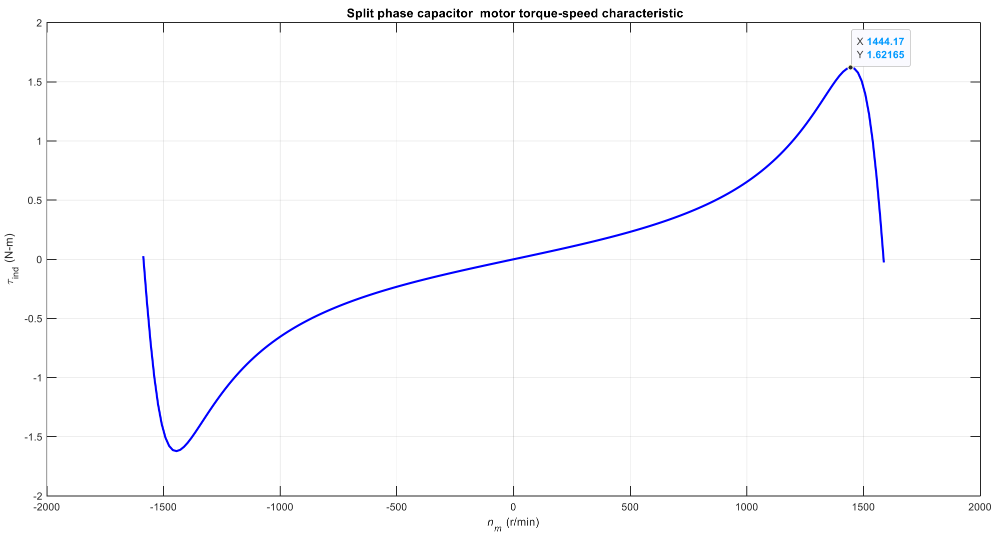


Figure: Torque Speed Characteristics

Maximum torque = 1.622

Matlab Code:

```
% M-file: prob9_4.m

% M-file create a plot of the torque-speed curve of the
% single-phase induction motor of Problem 9-4.
% First, initialize the values needed in this program.
r1 = 5.8; % Stator resistance
x1 = 24.04; % Stator reactance
r2 =6.67; % Rotor resistance
x2 = 24.04; % Rotor reactance
xm = 146.44% Magnetization branch reactance
v = 220; % Single-Phase voltage
n_sync = 1587; % Synchronous speed (r/min)
w_sync = 166.19; % Synchronous speed (rad/s)
% Specify slip ranges to plot
s = 0:0.01:2.0;
% Offset slips at 0 and 2 slightly to avoid divide by zero errors
s(1) = 0.0001;
s(201) = 1.9999;
% Get the corresponding speeds in rpm
nm = ( 1 - s) * n_sync;
% Caclulate Zf and Zb as a function of slip
zf = (r2 ./ s + j*x2) * (j*xm) ./ (r2 ./ s + j*x2 + j*xm);
zb = (r2 ./(2-s) + j*x2) * (j*xm) ./ (r2 ./(2-s) + j*x2 + j*xm);
i1 = v ./ ( r1 + j*x1 + 0.5*zf + 0.5*zb);
% Calculate the air-gap power
p_ag_f = abs(i1).^2 .* 0.5 .* real(zf);
p_ag_b = abs(i1).^2 .* 0.5 .* real(zb);
p_ag = p_ag_f - p_ag_b;
% Calculate torque in N-m.
t_ind = p_ag ./ w_sync;
% Plot the torque-speed curve
figure(1)
plot(nm,t_ind,'Color','b','LineWidth',2.0);
xlabel('\itn_{m} \rm(r/min)');
ylabel('\tau_{ind} \rm(N-m)');
title ('Split phase capacitor run motor torque-speed characteristic’, ‘FontSize’, 12);
grid on;
hold off;
```

Applications:

Capacitor-run motors are a specific type of single-phase induction motor that incorporates a capacitor in its circuit. This additional component enhances the motor's performance and makes it suitable for various applications. Here are some of the overall applications of capacitor-run motors in details-

- **Air Conditioning and HVAC Systems**: Capacitor-run motors are commonly used in air conditioning and heating systems for blower fans and condenser units. These motors provide the necessary starting torque to get the system running and can operate efficiently over extended periods.
- **Refrigeration Equipment**: Capacitor-run motors are found in various refrigeration systems, such as commercial refrigerators, freezers, and vending machines. They are efficient at maintaining constant temperature and can handle the variable loads of these cooling systems.
- **Compressors**: In small to medium-sized compressors for applications like air compressors, refrigeration compressors, and pneumatic tools, capacitor-run motors are employed for their ability to provide high starting torque and reliable operation.
- **Pumps**: Capacitor-run motors are used in various types of pumps, including water pumps for domestic and industrial purposes, pool pumps, and irrigation systems. Their ability to start and run under load makes them suitable for these applications.
- **Fans**: Ceiling fans, exhaust fans, and industrial fans often use capacitor-run motors due to their ability to provide the necessary torque for consistent and energy-efficient operation.
- **Conveyor Systems**: In manufacturing and material handling, conveyor systems benefit from capacitor-run motors as they can provide the initial thrust required to move heavy loads and maintain steady speed during operation.
- **Machine Tools**: Capacitor-run motors are used in various machine tools like lathes, milling machines, and drill presses, where they help provide the necessary power and precision for machining operations.
- **Blowers and Ventilation**: These motors are employed in blowers and ventilation systems, including those used in industrial settings, commercial kitchens, and residential exhaust systems.
- **Centrifugal Pumps**: Capacitor-run motors are well-suited for centrifugal pumps used in water supply systems, agricultural irrigation, and industrial processes, as they can handle varying loads efficiently.
- **Household Appliances**: Certain household appliances, such as washing machines, dryers, and dishwashers, use capacitor-run motors for their reliable and energy-efficient operation.

- ❖ **Automotive Applications**: In automotive settings, these motors can be found in devices like windshield wiper motors and fans used for cooling the engine or providing cabin ventilation.
- ❖ **Power Tools**: Some power tools, like circular saws and routers, use capacitor-run motors for their ability to provide high torque at startup and maintain consistent speed under load.
- ❖ **Agricultural Equipment**: Capacitor-run motors are used in various agricultural equipment, such as grain augers, feed mixers, and irrigation pumps, due to their ability to handle varying loads efficiently.

Cost Analysis:

No.	Instrument name	Cost (in Taka)
1	Motor	2700
2	Capacitor	80
3	Copper Wire, Fiber Paper, Insulation Tubes	1000
4	Transportation	350
5	Professional Assistance	500
6	Total	4630

A new motor costs over 5000 taka. So, our motor is cost efficient.

Summary:

A capacitor-run motor is a type of single-phase induction motor that includes a capacitor in its circuit. This capacitor provides an additional phase shift, enabling the motor to generate a starting torque, making it suitable for applications requiring high initial power. Capacitor-run motors find wide use in appliances like air conditioners, refrigeration units, pumps, fans, and various industrial machinery where reliable starting torque and steady-state operation are essential. Their versatility makes them indispensable in both residential and industrial settings.

In this project our main goal was to design a Capacitor-run motor. We not only have used the theoretical knowledge and calculations for choosing our design parameters but also have focused on fundamental ideas of designing a machine. Firstly, the functionality of the machine including where the motor will be used, how the motor will be used in various situations. The performance and cost analysis were a significant part of our design. In some parameters we choose performance over cost and for some others the opposite was true. Implementing the theoretical knowledge into practical engineering was not easy for us. We faced problems while implementing the designs, however, found out a suitable solution for that problem. We tried to design a motor with better performance but in a cost-effective way. We have also done the testing the motor for equivalent circuit parameters and the torque speed characteristics of our motor.

References

I. Electric Machinery Fundamentals (5TH Edition), Stephen J. Chapmen
McGraw-Hill International Edition
II. https://circuitglobe.com/capacitor-start-capacitor-run-motor.html
III. https://www.nidec.com/en/technology/motor/glossary/item/capacitor-run_motor/
IV. https://www.researchgate.net/figure/Capacitor-Run-Single-Phase-Induction-Motor_fig3_241590054
V. https://en.wikipedia.org/